Introduction to Computer Vision

Avinash Sharma

CSE Department, IIT Jodhpur

Image courtesy: https://www.etsy.com/in-en/listing/1607436850/third-eye-activation-spiritual-awakening

Past Affiliations

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY
HYDERABAD

UNIVERSITÉ DE GRENOBLE

Inria
INVENTEURS DU MONDE NUMÉRIQUE

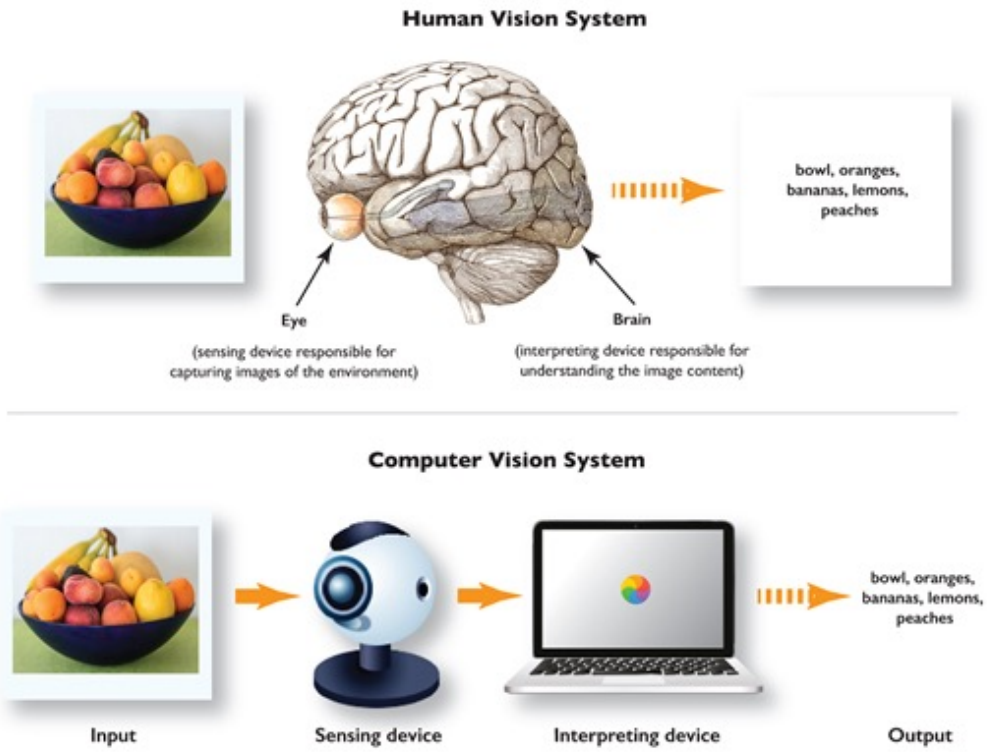
CVIT

LABORATOIRE JEAN KUNTZMANN
MATHÉMATIQUES APPLIQUÉES - INFORMATIQUE

mpii max planck institut informatik

xerox

What is Computer Vision ?



Courtesy: www.spiceworks.com

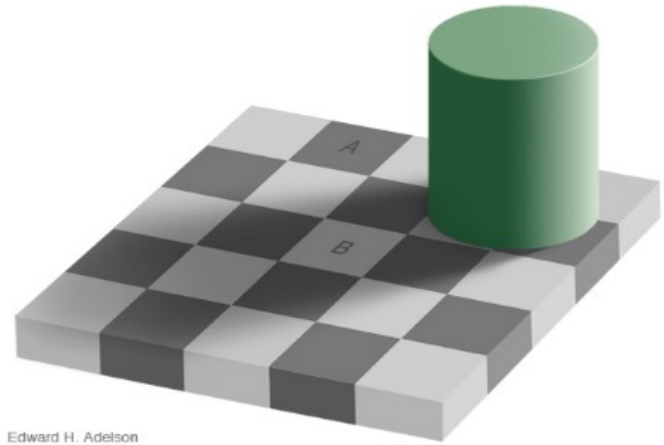
What is Computer Vision ?

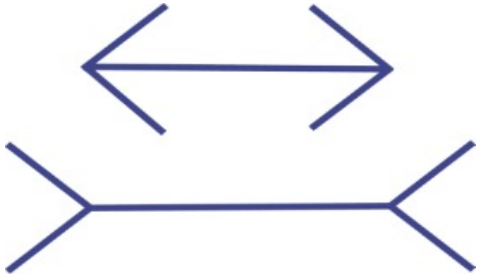
- Understanding of visual inputs (images/videos) by computers.
- Making sense out of them. Describing them.
- Does computer vision mimic the human vision?
  - Certainly, in many of its goals
  - Why? Human vision is among the best!
  - Sophisticated and efficient but not understood well
- Should computers process visual inputs like humans?

  Not necessarily!
  - Human visual system need not limit computer vision
  - We draw inspiration from it as often as is convenient

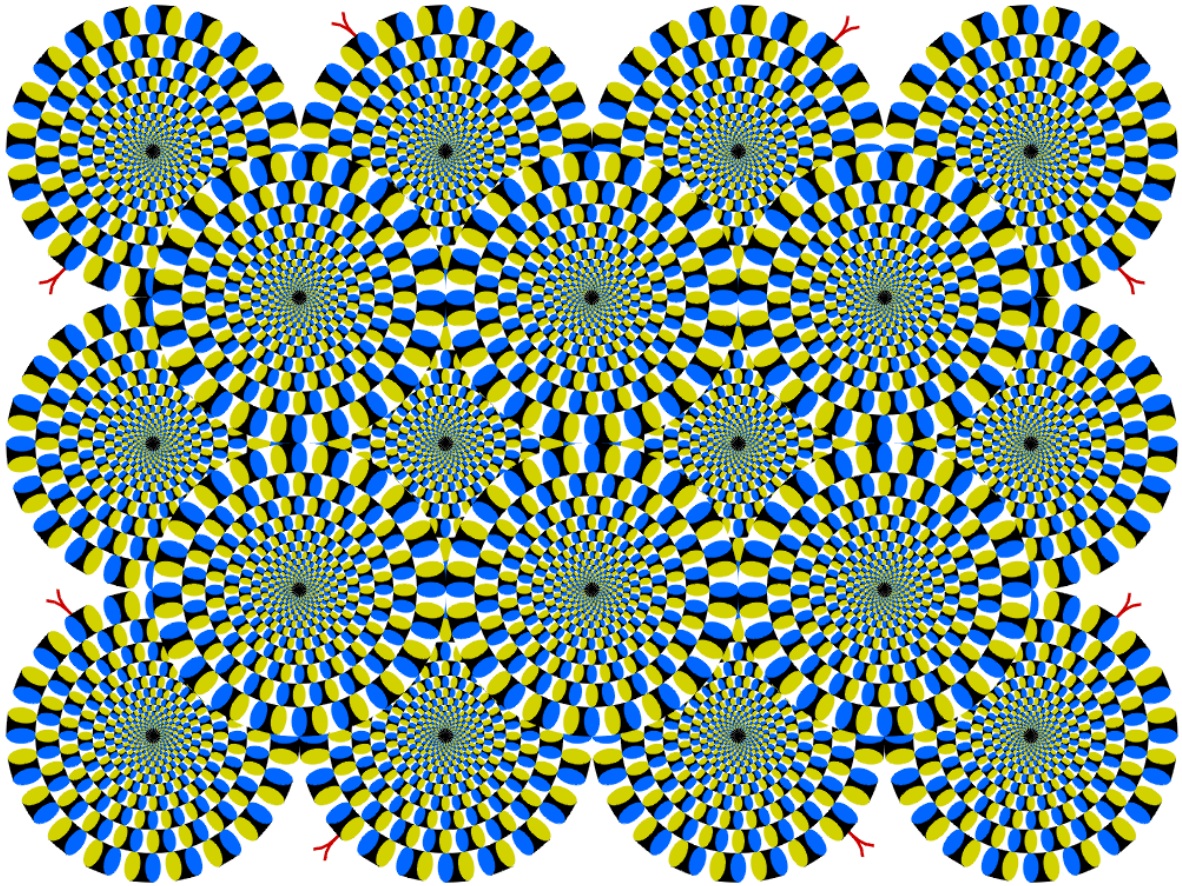
Limitations of Human Vision

Thatcher Effect

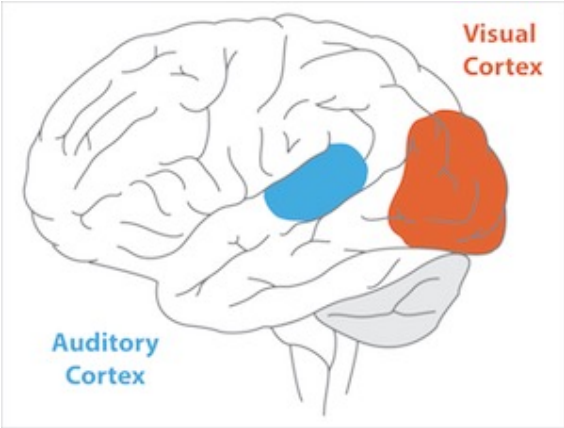




Limitations of Human Vision



Why Computer Vision?



- Visual cortex (large segment in brain)
- We perceive 3D world through eyes
- AI systems need similar capability for scaled deployment for task like driving, surveillance, navigation, etc.
- Lot of data 2D/3D visual data is generated by modern devices/systems

Deep Learning & Computer Vision !

- Deep Learning (DL) has become the primary driving force behind most recent success in CV
- Nevertheless, there is a lot to learn as part of classical Computer Vision with focus on mathematical and conceptual basis developed over 4 decades
  - Geometry
  - Optimization
  - Visual object representations
  - Optics, Lighting, Appearance models
- You need to know the basics to build on it

Course Plan

		Computer Vision - Sem1 2025 : Tentative Course Plan			
Lecture	Day	Date	Topic	Book/Chap	Assign./ Project
1	Th	31 Jul	Introduction, Image Formation	Gonsalve	
2	Fr	01 Aug	Image Processing: Filtering, Transformation, Color		
3	Tu	05 Aug	Image Processing: Spatial & Frequency Domain Filtering		
4	Th	07 Aug	Image Processing: Spatial & Frequency Domain Filtering		ASN 0
5	Fr	08 Aug	Camera Calibration		
6	Tu	12 Aug	2-View Geometry, Homography		
7	Th	14 Aug	2-View Geometry, Homography		ASN 1
8	Tu	19 Aug	Quiz 1		
9	Th	21 Aug	Stereo Correspondence, Depth from Stereo		
10	Fr	22 Aug	Stereo Correspondence, Depth from Stereo		
11	Tu	26 Aug	SFM / Bundle Adjustment		
12	Th	28 Aug	SFM / Bundle Adjustment		
13	Fr	29 Aug	Motion Estimation and Optical Flow		
14	Tu	02 Sep	Motion Estimation and Optical Flow		ASN 2
15	Th	04 Sep	Segmentation as Labelling: Introduction to Ncut		
16	Tu	09 Sep	Segmentation as Labelling: Introduction to Ncut		
17	Th	11 Sep	Image Segmentation by MRF		
18	Fr	12 Sep	Buffer		
19	Tu	16 Sep	Traditional Feature Detection & Description		
20	Tu	23 Sep	Traditional Feature Detection & Description		
21	Th	25 Sep	Traditional Feature Matching		ASN 3
22	Fr	26 Sep	Bag-of-Words Representation		
23	Tu	30 Sep	Quiz 2		
24	Fr	03 Oct	Intro to Neural-Nets (MLP)		
25	Mo	06 Oct	Intro to Conv-Neural-Nets (CNN)		
26	Tu	07 Oct	Intro to Conv-Neural-Nets (CNN)		
27	Th	09 Oct	CNNs for Detections		ASN 4
28	Fr	10 Oct	CNN for Recognition		Project Start
29	Tu	14 Oct	CNN for Semantic Segmentation		
30	Th	16 Oct	Recurrent NN for Video Analysis		
31	Fr	17 Oct	Generative Models for CV		
32	Tu	28 Oct	Generative Models for CV		
33	Th	30 Oct	Graph Neural Networks for CV Applications		
34	Fr	31 Oct	Vision Transformers (VT) for CV Applications		
35	Tu	04 Nov	3D Vision (Representation and Learning)		
36	Th	06 Nov	3D Vision (Representation and Learning)		
37	Tu	11 Nov	Applications: Novel View Synthesis		Submission
38	Th	13 Nov	Applications: 3D Terrain Modelling		
39	Fr	14 Nov	Applications: 3D Human Body Modelling		Viva
40	Tu	18 Nov	Selected Project Presentations		

Credit Distribution

Eval Heads	%Credits
Minor Exam	15
Major Exam	40
Quiz	5
Assignment	20 (5x4)
Course Project	20

Learning Resources

Resource	Author/Link
Digital Image Processing	Gonzales,Woods (pdf)
Multi-view Geometry in Computer Vision	Richard Hartley, Andrew Zisserman
Computer Vision A Modern Approach	Forsyth & Ponce
Lecture Slides	Selected slides **can** be shared !
Public Slides	Shared over GC
Video Lectures	NPTEL & other resources
Research Papers & Talks	Shared over GC

Guidelines

- 3 Key Steps to be followed:
  - Understand
  - Focus on ideas and insights
  - Implement
- No single textbook or resource.
- Continuous learning is desired.
- Aim ambitious course project.
- Any copy case in assignment or exam will be strictly penalized.

Knowing the Audience